DB-BlendMask: Decomposed Attention and Balanced BlendMask for Instance Segmentation of High-Resolution Remote Sensing Images

Zhenqian Chen, Yongheng Shang, *Member, IEEE*, Andre Python, Yuxiang Cai, and Jianwei Yin

***Abstract*— Instance segmentation is an important method for high-resolution remote sensing images (HRRSIs) analysis. Traditional instance segmentation algorithms are not suitable to analyze complex HRRSIs that exhibit: 1) various shapes and sizes of targets; 2) a large number of small targets; and 3) data with long tail distribution. Here we introduce DB-BlendMask, an efficient and accurate instance segmentation method that can accommodate complex HRRSIs. It is composed of size balance coefficient (SBC), class balance module (CBM), and decomposed attention blender module (DA-Blender module). SBC consists of a fair weight allocation strategy for positive samples in object detection. CBM combines classification obtained in object detection stage to guide the semantic feature extraction. Complementary to a traditional convolutional neural network (CNN) architecture, DA-Blender module has the ability to considerably compress space complexity of attention and merge attention with semantic feature to generate the instance mask. We compare the performance of DB-BlendMask with a benchmark Mask R-CNN on two typical datasets, iSAID, and ISPRS Postdam. We obtain an average detection precision of 39.2% on iSAID and 63.6% on ISPRS Postdam, which corresponds to an improvement of 2.5% and 2.7%, respectively, compared to the benchmark in a real-time scenario.**

***Index Terms*— BlendMask, class balance module (CBM), decomposed attention blender module (DA-Blender module), high-resolution remote sensing images (HRRSIs), instance segmentation, size balance coefficient (SBC).**

I. Introduction

THE rapid development of remote sensing technology has led to a growing increase in data collection and widely applied in environmental, territorial, and agricultural research areas with object detection [1]–[4] and semantic segmentation [5]–[9] techniques. To improve accuracy level and expand the remote sensing application, a wide range of instance segmentation algorithms on HRRSIs [10]–[12] have been researched. However, the detection ability of instance segmentation algorithm developed in computer vision (CV) is limited in the presence of [13]–[15] when applied in HRRSIs with the following challenges.

1) A large variety of shapes and sizes of targets. The shapes and sizes of ships, airplanes, and vehicles exhibit large variability in Fig. 1(b) compared to objects from natural views in Fig. 1(a).
2) A large number of small targets. The number of small targets (ships, vehicles) in HRRSIs is commonly larger compared to objects from natural views.
3) A long-tailed distribution of the data. The number of objects in one category [e.g., ships in Fig. 1(b1)] can exceed those observed in another category [e.g., ports in Fig. 1(a1)].

Instance segmentation consists of object detection and mask extraction. Common object detection methods in HRRSIs include anchor-based [1]–[3] and anchor-free [16]–[20] models. The effectiveness of anchor-based models mainly relies on the ratios and sizes of preset anchors. However, preset anchor boxes cannot accommodate geometrical complexities such as random direction, size variability and ratios of targets in HRRSIs. Anchor-free object detection overcomes the drawbacks in anchor-based models by directly locating targets by key point without requiring a preset box. However, setting



Fig. 1. Instances in natural images and HRRSIs. (a1) Ships, (a2) aircrafts, and (a3) vehicles in nature images. (b1) Ships, (b2) aircrafts, and (b3) vehicles in HRRSIs.

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the weight for numerous positive samples remains challenging. To solve this problem, Tian *et al.* [21] introduced FCOS, which computes centerness to evaluate the weight of positive samples. This approach ignores the deviation between weight and quality of the regression (intersection over union (IoU) [22] and generalized intersection over union (GIoU) [23]), which limits the domain of applications in HRRSIs. Traditional anchor-free object detection methods are not suitable to design a fair weight allocation strategy for targets with various shapes and sizes. Unfair weight allocation strategy for different positive samples leads to missing and reduplication of detection in small and large targets, respectively. To solve this problem, a fair weight strategy for size balance can be used to correctly estimate the bounding box location in HRRSIs.

Mask R-CNN [13] algorithm takes benefit from fully convolutional network (FCN) [24] to extract the mask in HRRSIs [10]–[12] following the selection of positive samples and their corresponding region of interest (RoI) by object detection. While the performance in extracting features may be similar to those obtained using semantic segmentation, the computational efficiency of FCN decreases in cases with a large number of small targets. In real cases, it is common to observe more than 1000 targets in HRRSIs [e.g., a large number of ships in a port, see Fig. 1(b1)], which exceeds the maximum number of targets (about 100) that can be accommodated in COCO [25]. To address this issue, the Blender module of BlendMask [26] utilizes matrix operation between attention and semantic feature of RoI to generate a mask. Blender module can achieve approximately $O(1)$ time complexity since it takes advantage of parallelism in GPU. Extra attention computation is required to execute this operation for each RoI results, which is computationally expensive, however. Another challenge in HRRSIs is the presence of long-tailed distribution in the data. This may lead to insufficient semantic feature representation for targets in minority categories. Current instance segmentation algorithms do not consider using the classification results obtained in object detection to guide semantic feature generation for each category, which leads to low-quality masks for targets in minority categories. To solve these issues, we consider both attention compression and context relationship between object detection and semantic segmentation, which allows us to accurately and efficiently generate masks.

In this article, we suggest **d**ecomposition attention and **b**alanced BlendMask (DB-BlendMask) shown in Fig. 2, a novel instance segmentation method that extends the domain of applications of the BlendMask framework. It includes a size balance coefficient (SBC) to correct centerness allocation strategy for targets with various sizes and shapes. Here the centerness branch in FCOS can infer the weight for positive samples and set a fair weight in loss computation of bounding box during training. In addition, the suggested approach includes a class balance module (CBM), which utilizes classification as prior knowledge to guarantee the sufficient semantic feature representation for each category, even in presence of long-tailed distribution. To accommodate the presence of a large number of small targets a decomposition attention blender (DA-Blender) module is used in alternative to the traditional FCN based on matrix operation in instance segmentation to achieve $O(1)$ time complexity by compressing the number of parameters in attention. The main contributions of our work are as follows.

1) Introduction of a novel instance segmentation approach, DB-BlendMask, which uses SBC, CBM, and a decomposition attention blender module (DA-Blender module) to improve accuracy and speed in the analysis of complex HRRSIs.
2) SBC allocates fair weight for positive samples in targets with various sizes during training. Compared with other weight optimization approaches used to identify RoIs, SBC accounts for size variability in positive samples, which avoids missing and reduplication detection in small and large targets, respectively.
3) We design CBM to aggregate the instance classification and spatial information to strengthen the feature representation for each category. Compared with the classification attention in semantic segmentation, CBM directly utilizes classification in object detection, without constructing an additional complex attention module.
4) DA-Blender module maintains approximately $O(1)$ in time complexity during inference and compresses the spatial complexity of attention from $O(n^2)$ to $O(n)$, which contributes to reduce the number of parameters while ensuring relatively high accuracy and efficiency.

II. Related Work

Since instance segmentation contains object detection and semantic segmentation, we review three distinct sections of the literature: Object detection, semantic segmentation, and instance segmentation.

A. Object Detection

Early object detection methods [27]–[29] have mainly relied on prior knowledge on handcrafted characteristics. Deep learning algorithms such as anchor-based [30]–[34] and anchor-free [21], [35]–[37] models have been widely employed in the detection of sewage treatment plants [2], garbage [16], ships [18], and aircrafts [20] in HRRSIs. Anchor-based representative networks can be based on R-CNN series [30]–[32], Yolo series [33], [34], RetinaNet [38], and FPN [39]. They set up anchors of fixed shape and adjust them during the regression process. Faster R-CNN [32] approaches design a region proposal network and directly regress targets by extracting features. FPN [39] utilizes a feature pyramid structure to capture targets with different scales based on Faster R-CNN. Inspired by anchor-based idea, PBNet [2] utilizes part information into context information to improve the performance in the detection of composite objects. Dong *et al.* [40] used suitable object scale features to obtain adaptive RoI scale in aircraft detection.

In contrast to anchor-based methods, anchor-free algorithms do not need to preset anchor boxes. Based on different resolution features generated on FPN, anchor-free algorithms complete the regression process by key points and present several variations. FCOS [21] computes centerness to distinguish the quality of positive samples. ATSS [41] further discusses the situation of redundant positive samples in large targets on the basis of FCOS. Extensions of anchor-based models have been

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proposed to detect objects in HRRSIs. Shi *et al.* [19] expanded the calculation of centerness into a form similar to focal loss [38] and took advantage of the symmetric characteristics of the object. Wei *et al.* [20] formulated the aircraft detection task as prediction and clustering of paired intersecting line segments. Despite the important progress made by anchor-free models, the distinction of the quality of positive samples remains challenging for this class of models. Ineluctably, incorrect weight allocation for positive samples leads to poor detection performance, such as missing and reduplication detection.

B. Semantic Segmentation

Semantic segmentation is an expansion of image classification, which aims to predict the classification of each pixel in the image. FCN [24], the first generation of semantic segmentation models, realizes an end-to-end pixel label prediction through full convolution network. However, the extraction of feature representation remains challenging for small targets. The encoder-decoder structures [42]–[44] mitigate this issue by aggregating the location and context information to extract the characteristics of targets represented at different scales. In HRRSIs, DABNet [5] achieves high-efficiency in cloud detection by using a light network with a deformable context feature pyramid module and boundary-weighted loss. Tree-UNet [45] adopts a Tree-CNN block structure and transmits feature maps through concatenating connections in ISPRS dataset [46]. Models with an encoder-decoder structure cannot effectively build the category and channel relationship between targets, however. To improve the performance in feature extraction [7], [9], Niu *et al.* [7] designed an efficient attention-based framework for adaptive capture of global dependencies from spatial, channel and category perspectives. Mou *et al.* [9] proposed spatial relation and channel relation modules to learn and infer the global relationship between any two spatial positions or feature maps. Despite the advances made by encoder-decoder models, mask segmentation for RoIs remains a costly process in instance segmentation. These methods for semantic segmentation are computationally impractical in the presence of HRRSIs that exhibit numerous targets.

C. Instance Segmentation

Instance segmentation can locate targets by instance and pixel simultaneously. Building on the Faster R-CNN architecture, pioneering instance algorithm Mask R-CNN [13] extends the commonly-used RoIPooling approach with RoIAlign and adds the FCN layer. Further works have also been carried out on improving feature extraction [14] and the design of alternative loss functions [15] to improve the detection performance of Mask R-CNN. In HRRSIs, Pan *et al.* [10] proposed rotated instead of horizontal boxes, which improves the extraction of RoIs. Xu *et al.* [47] designed a gated spatial memory and centroid-aware network (GSMC) to address issues associated with the presence of a complex background and a large variety in building appearance. Object detection architecture of anchor-free instance segmentation algorithms have also been recently enhanced [26], [48], [49]. Based on anchor-free object detection FCOS, CenterMask [48] discusses the application of anchor-free instance segmentation in HRRSIs. Liu *et al.* [50] designed an algorithm that improves the reliability of feature fusion in FPN. Wu *et al.* [51] extends the domain of application of feature extraction to spatial attention module. To address the effects associated with the complexity of FCN for RoIs, BlendMask [26] suggests a blender module to efficiently merge attention and feature for mask generation.

The structure of Mask R-CNN, CenterMask and BlendMask can be summarized as follows.

1) Mask R-CNN = Faster R-CNN + FCN.
2) CenterMask = FCOS + FCN.
3) BlendMask = FCOS + Blender.

So far, the literature has mainly focused on context information enhancement after FPN or mask generation without considering computational efficiency and the number of parameters into consideration, which has limited the domain of potential applications in HRRSIs.

III. METHODOLOGY

We suggest DB-BlendMask that constructs an accuracy-efficiency balance instance segmentation network through training optimization, context information enhancement, and attention parameters compression. First, we introduce the weight allocation process of SBC. Second, we discuss spatial attention and classification combination in semantic feature generation in CBM. Third, we describe the DA-Blender module, which decomposes the attention and merges semantic feature with attention to generate a mask for each RoI. Finally, we discuss the design of the multitask loss function used in DB-BlendMask.

A. Size Balance Coefficient

Box sub-branch in FCOS regresses the 4D vector $\boldsymbol{t}^* = (l^*, b^*, r^*, t^*)$ for each location (x, y) shown in Fig. 3(a) and (d), where $\boldsymbol{t}^* = (l^*, b^*, r^*, t^*)$ is the distance from (x, y) to each edge of box. During training, FCOS uses centerness in (1) to evaluate the weight of regression loss to suppress the positive samples that deviate substantially from the center. During the inference process, FCOS predicts centerness $o_{x,y}$ in centerness branch as a weight of classification score $\boldsymbol{p_{x,y}}$ as shown in (2)

$$\text{centerness}^* = \sqrt{\frac{\min(l^*, r^*)}{\max(l^*, r^*)} \times \frac{\min(t^*, b^*)}{\max(t^*, b^*)}} \tag{1}$$

$$\boldsymbol{s_{x,y}} = \sqrt{\boldsymbol{p_{x,y}} \times o_{x,y}}. \tag{2}$$

However, deviation is observed in both centerness and the quality of the bounding box regression during training. Despite the large GIoU values obtained in the positive samples [23] with ground truth [Fig. 3(b)], the contribution is suppressed by centerness given the large deviation with the center of ground truth. In Fig. 3(d), large objects with four positive samples exhibit a higher probability to get the large values of centerness and GIoU. Fig. 3(e) suggests a good fit (orange and red ones) between the results of the regression model and the ground truth. The seemingly bad fit (green and blue ones) associated with a small GIoU indicates redundant loss.

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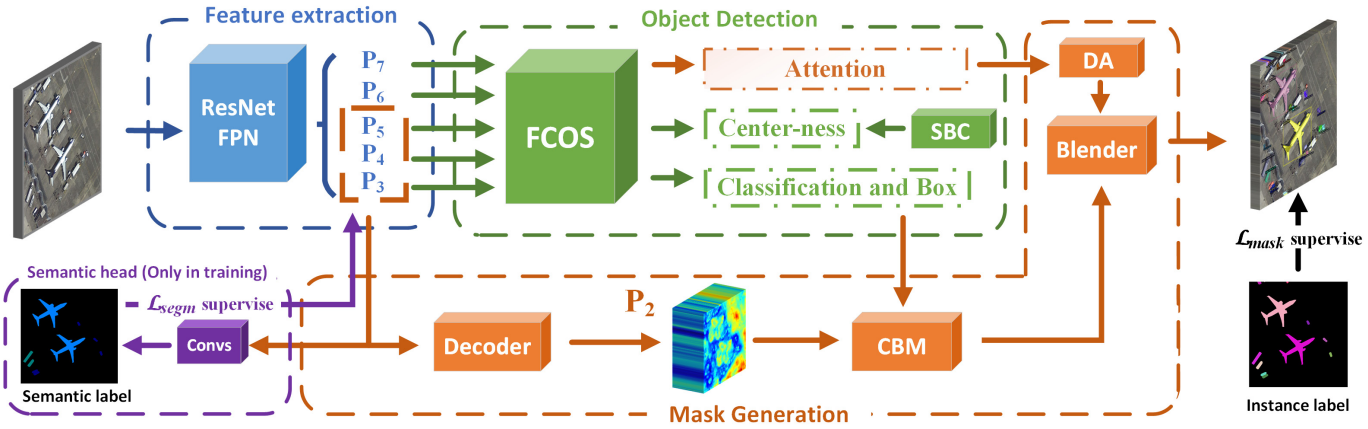


Fig. 2. DB-BlendMask overall framework. The input image first passes through the feature extraction module (blue) to generate features at different resolutions. The extracted features are fed into the object detection head (green) FCOS to acquire RoI information, where SBC is used to correctly allocate centerness and DA decomposes the attention. Meanwhile, features of different resolutions are fed into the decoder module to generate semantic feature P_2. CBM combines P_2 with classification and anchor box in FCOS to gather semantic information for each RoI. Finally, the blender merges the semantic feature of RoI with attention to obtain instance masks. The semantic head (purple) uses a semantic label to supervise the generation of feature with $\mathcal{L}_{\text{mask}}$ during training.

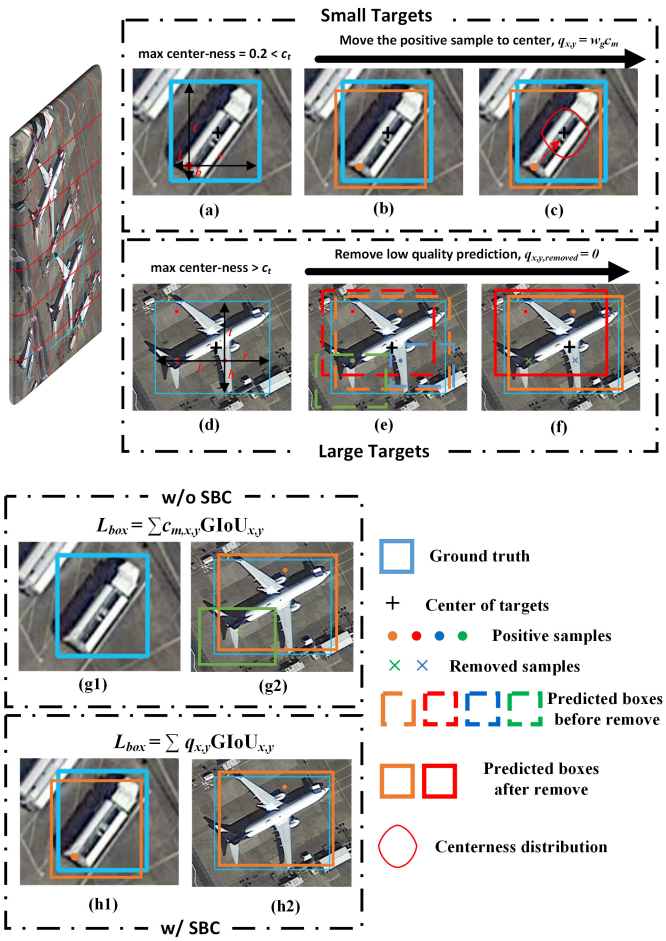


Fig. 3. Main processes involved in SBC. (a) Ground truth of small targets. (b) Predicted boxes in small targets. (c) Centerness threshold distribution. (d) Ground truth of large targets. (e) Predicted boxes in large targets. (f) Retained boxes. (g) Inference results without SBC. (h) Inference results with SBC.

During inference, we identify background as a score $s_{x,y}$ of points less than a threshold. An underestimation of the predicted weights by the centerness branch for small targets results in the detection of missing, as illustrated in Fig. 3(g1). On the contrary, redundant positive samples of large targets with high weights occupying the top-k scores for NMS [52] is likely to cause more than one bounding box for a target [see Fig. 3(g2)].

To avoid unfair weight allocation in small and large targets, we design SBC, expressed in Algorithm 1. The input includes a set of ground truth $\mathcal{G}$ and a set of positive samples $\mathcal{A}$. The output Q is the coefficient for corresponding samples in $\mathcal{A}$.

1) Global Object Information Extraction: We compute the average number of positive samples for each ground truth μ_{pa} to evaluate the average size of targets in one image. Then, we acquire centerness threshold c_t, which is used to judge whether the positive sample is the extreme condition of weight under suppression (Algorithm 1, lines 1–7).

2) Ground Truth Information Extraction: We traverse each ground truth g in $\mathcal{G}$ to calculate statistics for positive samples in ground truth g (Algorithm 1, lines 8–17). We obtain w_g inversely proportional to the ground truth size N_g, which means that smaller targets are assigned a larger weight during training (Algorithm 1, line 12). We compute centerness $c_{\max}$, which is used to identify extreme condition [Fig. 3(b), Algorithm 1, line 13]. After the realization of these steps, we compute the GIoU threshold t_g to filter out the low-quality predictions whose GIoU less than t_g.

3) Positive Samples Coefficient Setting: We traverse positive samples in ground truth g. If $c_{\max} < c_t$, we define positive samples in this ground truth as extreme condition and we set the weight for these positive samples as the max value between weighted centerness c_w and centerness threshold c_t. The effect is equal to taking candidates into centerness distribution, as illustrated in Fig. 3(c). Otherwise, samples whose GIoU value $d_m < t_g$ are set to zero and removed during training [see Fig. 3(f) and Algorithm 1 (lines 18–31)]. Note that weights associated with left samples keep their original centerness.

Compared with other weight allocation methods such as the ATSS [41], which only mitigates issues associated with low-quality positive samples in large targets, SBC considers

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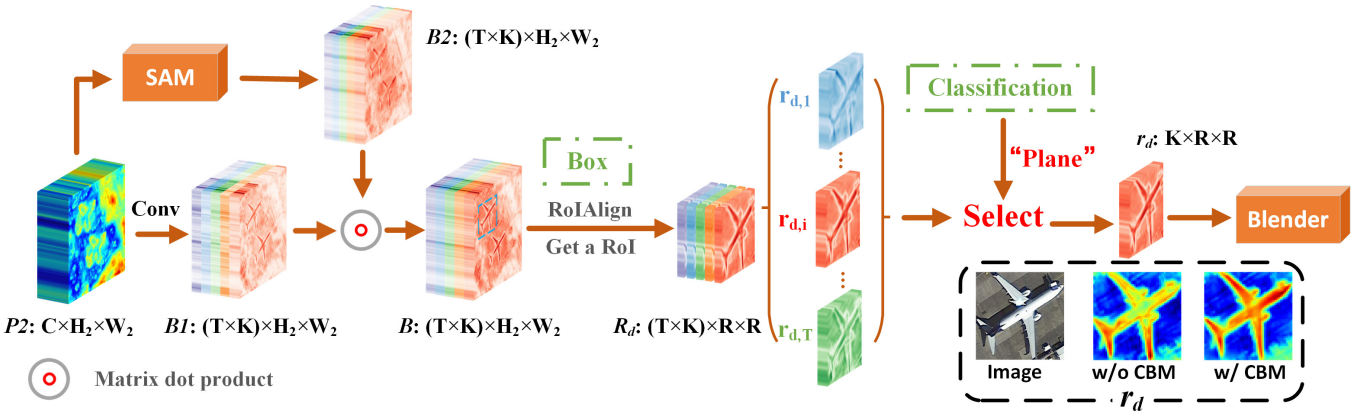


Fig. 4. Architecture of CBM. It first applies convolution layer and spatial attention module. After executing the RoIAlign process, features with K channels, the object detection procedure selects the corresponding RoI r_d for blender module. Colors represent each category and classification obtained during the process (e.g., red represents plane, blue represents vehicles, green represents ship and classification selects its corresponding category for r_d).

positive samples in all conditions. SBC can assign correct weights for positive samples, as illustrated in Fig. 3(h). In addition, this process does not need changes on the architecture of the network in training or extra computation during the inference process.

B. Class Balance Module

CBM includes SAM introduced in CenterMask [48] and classification selection (CS) (Fig. 4). After obtaining $P_2 \in \mathbb{R}^{N\times C\times H_2\times W_2}$ from P_3, P_4 and P_5 by Deeplab V3+ decoder [44], we use convolution and SAM to acquire $B_1, B_2 \in \mathbb{R}^{N\times(T\times K)\times H_2\times W_2}$, where N is batch size, H_2, W_2 set the resolution of the second layer in FPN and C is the channel number of P_2. The number of masks for semantic feature and attention is given by the parameter K. T is the number of categories represented by colors in $B, B1, B2, R_d$. SAM can effectively extract key information from the channel of features. It first employs global average and max pooling, respectively, to get the value by channel. Then SAM concatenates the max and average features $P_{\max}, P_{\text{avg}} \in \mathbb{R}^{N\times 1\times H_2\times W_2}$, applies a convolution layer and sigmoid function to get $B_2 \in \mathbb{R}^{N\times(T\times K)\times H_2\times W_2}$ (3). After these steps, we compute the dot product B_1 and B_2 to obtain the final base feature $B \in \mathbb{R}^{N\times(T\times K)\times H_2\times W_2}$

$$B_2 = \sigma(\mathcal{F}_{3\times 3}(\text{Concat}(P_{\max}, P_{\text{avg}}))). \tag{3}$$

We use RoIAlign to get the set of RoIs $\boldsymbol{R'} = \{r'_d \in \mathbb{R}^{T\times K\times R\times R} | d = 1, \ldots, D\}$ (4), where D is the number of RoIs and d is the index in set $\boldsymbol{R'}$

$$r'_d = \text{RoIAlign}_{R\times R}(B, p_d) \quad \forall d \in \{1, \ldots, D\}. \tag{4}$$

The classification set of RoIs $\boldsymbol{Cls} = \{c_d \in \mathbb{R}^{1\times T} | d = 1, \ldots, D\}$ is predicted as a one-hot label in FCOS, where the highest probability is 1 and others are set to zero. This operation allows us to select the semantic feature $\boldsymbol{R} = \{r_d \in \mathbb{R}^{K\times R\times R} | d = 1, \ldots, D\}$ for each corresponding category by computing the matrix product (5)

$$r_d = c_d \times r'_d \quad \forall d \in \{1, \ldots, D\}. \tag{5}$$

Compared with the mask generation module in BlendMask [26], CBM can accommodate spatial attention with SAM and uses classification to select the corresponding semantic feature of its category. The computation of r_d (Fig. 4) suggests that CBM provides sufficient accuracy in representing features to account for data affected by long-tailed distribution in different categories, such as targets in minority categories (aircraft versus vehicles in this example).

C. DA-Blender Module

Decomposition attention (DA) provides an effective and flexible attention representation, which reduces the computational burden while improving accuracy. For a cropped image with a 800 × 800 resolution in iSAID [53], there are 13 343 ($100^2+50^2+25^2+13^2+7^2$) samples in total from FPN levels (P_3, P_4, P_5, P_6, P_7). For input attention $\in \mathbb{R}^{H_l\times W_l\times K\times M\times M}$, each point has the attention with $K \times M \times M$ size, where $K = 4$ is the channel number and $M = 14$ is the resolution of attention. H_l and W_l define the resolution used for each corresponding FPN level. The number of attention parameters is almost 10 million (13 343 × 4 × 14 × 14) and occupies one third of the total of BlendMask. Previous results obtained using BlendMask [26] indicate that a larger resolution M can account for more details of the instance to achieve higher precision, however the process is restricted by the space cost of attention. Since the space complexity is $O(M^2)$, M is a key hyper-parameter to optimize.

Fig. 5 indicates that RoI r_d contributes more in instance details when the visualization of attention s_d is blurry. It suggests that r_d is likely to play a main role in mask segmentation and representation of attention by pixel is redundant. DA designs a m-to-n projection, where m is considered smaller than n to decrease the memory cost by matrix decomposition.

In DA, we use a constant L less than M and $K \times M \times L$ to replace the original attention size $K \times M \times M$ (Fig. 6). First, we split the attention into $a_{d,\text{I}}, a_{d,\text{II}} \in \mathbb{R}^{K\times M\times L/2}$ and then we transpose $a_{d,\text{II}}$ into $a^{\text{T}}_{d,\text{II}} \in \mathbb{R}^{K\times L/2\times M}$. Finally, we batch matrix multiply $a_{d,\text{I}}$ and $a^{\text{T}}_{d,\text{II}}$ to achieve $a_{d,\text{out}} \in \mathbb{R}^{K\times M\times M}$ (6)

$$a_{d,\text{out}} = a_{d,\text{I}} \times a^{\text{T}}_{d,\text{II}} \quad \forall d \in \{1, \ldots, D\}. \tag{6}$$

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Algorithm 1 SBC

Input:
$\mathcal{G}$: set of ground-truth boxes
$\mathcal{A}$: set of positive samples
Output:
Q: array of coefficient for all positive samples

$N_a \leftarrow$ number of ground-truth
$N_p \leftarrow$ number of positive samples
average number of positive samples for each ground-truth $\mu_{pa} = \frac{N_p}{N_a}$
centerness of all positive samples with ground-truth $\mathcal{C} = Centerness(\mathcal{A}, \mathcal{G})$;
mean of $\mathcal{C} : \mu_c = Mean(\mathcal{C})$;
standard deviation of $\mathcal{C} : \sigma_c = Std(\mathcal{C})$
centerness threshold for $\mathcal{A} : c_t = \mu_c + \lambda_c \sigma_c$
for each ground-truth $g \in \mathcal{G}$ **do**
  $\mathcal{M}_g \leftarrow$ select positive samples for ground-truth g;
  $\mathcal{C}_g \leftarrow$ select centerness from $\mathcal{C}$ for ground-truth g;
  $N_g \leftarrow$ number of positive samples in $\mathcal{M}_g$;
  $w_g = \sqrt{\frac{\mu_{pa}}{l_m}}$;
  max centerness of $\mathcal{C}_g : c_{max} = \max(\mathcal{C}_g)$;
  GIoU between $\mathcal{M}_g$ and $g : \mathcal{D}_g = GIoU(\mathcal{M}_g, g)$;
  $\mu_g \leftarrow$ mean of $\mathcal{D}_g$;
  $\sigma_g \leftarrow$ standard deviation of $\mathcal{D}_g$;
  GIoU threshold for ground-truth $g : t_g = \mu_g + \lambda_g \sigma_g$;
  for each candidate $m \in \mathcal{M}_g$ **do**
    $id_m \leftarrow id$ of m in positive samples set $\mathcal{A}$
    $c_m = Centerness(m, g)$;
    $c_w = w_g c_m$;
    $d_m = GIoU(m, g)$;
    if $c_{max} < c_t$ **then**
      $Q[id_m] = \max(c_w, c_t)$;
    else
      if $d_m < t_g$ **then**
        $Q[id_m] = 0$;
      else
        $Q[id_m] = c_m$;
      end
    end
  end
end
return Q;

Since the resolution in attention a_d is M smaller than R, we apply a bilinear interpolation method to extend the resolution of a_d from M to R. Then a'_d is normalized with a softmax function along K dimensions to obtain a set of score maps s_d.

We illustrate the process by considering a mask generation process of a typical airplane. Fig. 5 shows four images associated with the attentions s_d and RoIs r_d for $K = 4$. Then we apply element-wise product between each entity r_d and s_d sum along the K dimension using a sigmoid function to generate a mask m_d (7). In Fig. 5, colors indicate k semantic

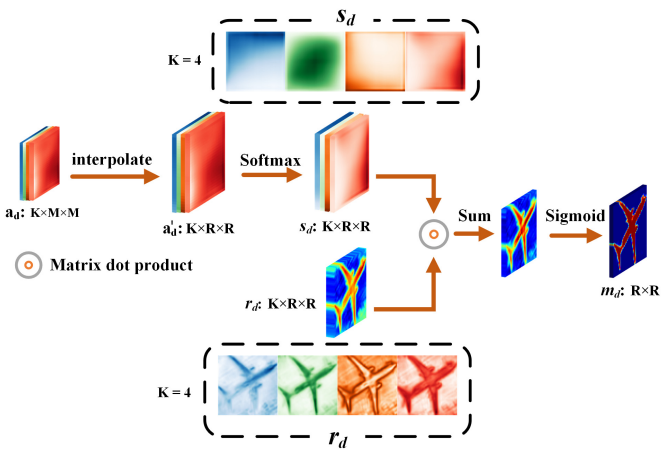


Fig. 5. Architecture of blender module. Different colors representing corresponding channels in r_d learn the different feature of RoI (e.g., when $K = 4$, first blue channel learns the head and the wing of the aircraft. Second green channel represents the inner feature of the aircraft. Third orange channel extracts the edge feature of the aircraft and the last red channel focuses on the tail and right wing of the aircraft).

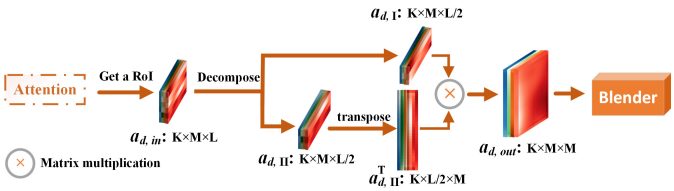


Fig. 6. Architecture of DA.

feature and attention indices

$$m_d = \text{Sigmod}\left(\sum_{k=1}^{K} s_{d,k} \circ r_{d,k}\right) \quad \forall d \in \{1, \ldots, D\}. \tag{7}$$

DA-Blender compresses the space complexity of attention from $O(n^2)$ to $O(n)$ and releases a limitation for attention size M. In contrast to FCN, DA-Blender executes matrix operations between attention and the feature matrix to generate mask. This approach reduces the computation burden that may occur with a large number of instances. In Section IV, our experiments show that precision remains stable when the floating-point operations per second (FLOPs) and number of parameters decrease. The saved space can be used to set a larger M to account for more details of each instance and improve the precision.

D. Training Loss

We define our training loss function in (8), where $\mathcal{L}_{\text{cls}}$ is the focal loss for classification as in [38], $\mathcal{L}_{\text{box}}$ is the GIoU loss as in [23] for bounding box regression and $\mathcal{L}_{\text{ctr}}$ is the cross entropy loss for centerness. N_p denotes the number of positive samples and $\mathbb{1}_{\{c^*_{x,y}>0, q^*_{x,y}>0\}}$ is the indicator function, being 1 if category index $c^*_{x,y} > 0$ and weight adjusted by SBC $q^*_{x,y} > 0$, or 0 otherwise. $\boldsymbol{p}_{x,y}$ is the prediction of classification score vector for each category and $\boldsymbol{t}_{x,y}$ is the regression prediction. $q_{x,y}$ is the weight for classification predicted by centerness

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branch

$$\begin{aligned}\mathcal{L}(\{\boldsymbol{p}_{x,y}\},\{\boldsymbol{t}_{x,y}\},\{q_{x,y}\}) &= \frac{1}{N_p}\sum_{x,y}\mathcal{L}_{\text{cls}}\left(\boldsymbol{p}_{x,y},c^*_{x,y}\right) \\ &+ \frac{\lambda_{\text{box}}}{N_p}\sum_{x,y}\mathbb{1}_{\{c^*_{x,y}>0,q^*_{x,y}>0\}}q^*_{x,y}\mathcal{L}_{\text{box}}\left(\boldsymbol{t}_{x,y},\boldsymbol{t}^*_{x,y}\right) \\ &+ \frac{\lambda_{\text{ctr}}}{N_p}\sum_{x,y}\mathbb{1}_{\{c^*_{x,y}>0,q^*_{x,y}>0\}}\mathcal{L}_{\text{ctr}}(q_{x,y},q^*_{x,y}) \\ &+ \frac{\lambda_{\text{segm}}}{H_2W_2}\sum_{x,y}\mathcal{L}_{\text{segm}}\left(\boldsymbol{p}_{x,y},c^*_{x,y}\right) \\ &+ \frac{\lambda_{\text{mask}}}{N_r}\sum_{x,y}\mathcal{L}_{\text{mask}}\left(\boldsymbol{p}_{x,y},c^*_{x,y}\right). \end{aligned} \tag{8}$$

$\mathcal{L}_{\text{segm}}$ and $\mathcal{L}_{\text{mask}}$ are cross entropy losses associated with the mask generation shown in Fig. 2. After obtaining P_2 by DeeplabV3+ decoder, we apply three convolution layers to generate the semantic segmentation result to supervise the feature extraction [26] in semantic head (see Fig. 2). H_2 and W_2 set the resolution of P_2. $\mathcal{L}_{\text{mask}}$ is the loss for instance mask generation and N_r is the number of pixels in RoIs. The hyperparameter λ_{box}, λ_{ctr}, λ_{mask}, and λ_{segm} control the trade-off of the loss. Here we set $\lambda_{\text{segm}} = 0.3$ and the other hyper-parameters as 1. We further discuss the influence of changes on values of λ_{segm} in Section IV.

IV. Experimental Results and Discussion

A. Datasets

1) *iSAID:* Most images in iSAID are collected from Google Earth. Additional images are obtained from the Chinese satellites JL-1 and GF-2. iSAID is the extension of [54] annotated at pixel level and contains 15 categories. The total number of instances is 655451 across 2806 HRRSIs. We crop the HRRSIs on the patches of resolution 800×800 with a stride set to 200. Detailed information about training, validation, test division, and number of each instance after crop is provided in Table I.
2) *ISPRS Postdam:* ISPRS Postdam [46] is the semantic segmentation dataset of HRRSIs in urban areas and the resolution of characteristic image data is 8 meters. It includes 38 remote sensing images, and the bands are red, green, blue, and near-infrared. In our experiment, we keep all RGB channels. Six categories are annotated including impervious surfaces, building, low vegetation, tree, car, and background. Since the annotation is for semantic segmentation, we translate the dataset into instance segmentation set by distinguishing the contours. However, this transferring method leads to considerable deviations for road and trees since objects in these categories are difficult to distinguish. Therefore, we only save buildings and cars for evaluation and consider other categories as background with based on the same preprocessing operated in iSAID. Detailed information is provided in Table II.

TABLE I
INSTANCE NUMBER DISTRIBUTION OF TRAINING, VALIDATION, AND TEST SET IN iSAID

	Class Name	Train	Val	Test	Sum
1	Plane	18457	6447		24904
2	Baseball diamond(BD)	1124	473		1597
3	Bridge	4459	1088		5547
4	Ground field track(GTF)	1176	456		1632
5	Small vehicle(SV)	474718	161908		636626
6	Large vehicle(LV)	80734	19861		100595
7	Ship	79180	28304		107484
8	Tennis court(TC)	6449	2017	-	8466
9	Basketball court(BC)	1490	409		1899
10	Storage tank(ST)	13019	6956		19975
11	Soccer-ball field(SBF)	1571	622		2193
12	Roundabout(RA)	955	377		1332
13	Harbor	15006	4995		20001
14	Swimming pool(SP)	4909	1412		6321
15	Helicopter(HC)	1437	206		1643
Total		704684	235531	-	940215

TABLE II
INSTANCE NUMBER DISTRIBUTION OF TRAINING, VALIDATION, AND TEST SET IN ISPRS POSTDAM

	Class Name	Train	Val	Test	Sum
1	Building	3873	698	1908	6479
2	Car	11259	2084	5415	18758
Total		15132	2782	7323	25237

B. Evaluation Metrics

We use average precision (AP) and AP of bounding box (APbb) to evaluate the instance segmentation and object detection performance. Besides, we use average detection time to evaluate the efficiency. AP and APbb are the area of the precision-recall curve (PRC). This allows us to assess the ability of the model to detect positive cases when the data are imbalanced [55]. Precision and recall are can be obtained from elements of the confusion matrix, which consists of the number of true positives (TP), false positives (FP) and false negatives (FN). Precision is the fraction of positive instances among all instances predicted as positives and can be expressed as TP/(TP + FP). Recall (also called sensitivity) measures the ability of the model to find all relevant cases in the data and can be expressed as TP/(TP + FN) [56].

C. Implementation Details

Since high scale variation is commonly observed in aerial images (e.g., playground versus small vehicles) from iSAID, we use multiscale augmentations at six scales $(400, 600, 800, 1000, 1200)$ and set the longer edge as 1400 pixels. Compared with no argumentation and default setting in COCO [25], the results of BlendMask (Table III) indicate that high variation multiscale argumentation can considerably enhance the extraction of targets at different scales and improve the performance of the detection process.

Fig. 2 suggests that semantic generation supervised by $\mathcal{L}_{\text{segm}}$ plays an auxiliary role to enhance the representation of P_2 when $\mathcal{L}_{\text{mask}}$ is used to supervise the mask for each instance. Large λ_{segm} in multitask regression seem to influence the loss contribution of others. The results in Table IV indicate that

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TABLE III
DIFFERENT MULTISCALE ARGUMENTATION IN TRAINING

Scale	AP	AP_{bb}
(800)	33.30	39.85
(640, 672, 704, 736, 768, 800) [25]	34.40	40.59
(400, 600, 800, 1000, 1200)	**37.82**	**44.83**

TABLE IV
EXPERIMENT TO DETERMINE THE WEIGHT OF λ_{SEGM} IN THE VALIDATION DATASET OF iSAID

λ_{segm}	AP	AP_{50}	AP^{bb}	AP^{bb}_{50}
0	37.3	61.9	44.7	66.1
0.3	**37.8**	**62.3**	**44.8**	**66.3**
1	37.2	61.6	43.9	65.7

$\lambda_{\text{segm}} = 0.3$ is an appropriate setting for auxiliary enhancement as summarized in Table IV.

We set the threshold of nonmaximum suppression (NMS) as 0.6 (default value is 0.5) and set the number of detection boxes as 1000 (default value is 100) in COCO during evaluation. For ISPRS Postdam, we keep the default set in [13] and [26]. More details on the hyperparameters setting is provided in Table V, which highlights different settings for iSAID and ISPRS Postdam.

All experiments are conducted on a server with 48 kernels Intel(R) Xeon(R) Silver 4214 CPU, 64 GB RAM, and 2 Telsa P40 GPUs. All models are trained on two GPUs with batch size 6. The speed is tested on single GPU P40 during inference. We do the ablation experiments on the validation dataset of iSAID and compare the results with other models on the test set of iSAID and ISPRS Postdam. We use ResNet-50 [57] as backbone and train the model using 90 k iterations with an initial learning rate of 0.025, which is then decreased by a factor of 10 at 60 k and 80 k iteration. In the test set, models are trained with 180 k iterations and the decrease learning rate set at 120 k and 160 k decreases by a factor of 10 with the same learning rate.

D. Results and Analysis

1) Analysis of the Coefficient SBC: The λ_c represents the trade-off of centerness standard deviation for each image in SBC. A larger λ_c indicates that more ground truth will be considered as extreme samples and the centerness value will be weighted when lower λ_c performs inversely. If λ_c reaches $+\infty$, we assume that all ground truth are extreme samples. As shown in Table VI, we set λ_c in $(-0.50, -0.25, 0, 0.25, 0.50)$ and use $\lambda_c = -0.25$, which shows the best performance both in 38.0% AP and 44.9% AP^{bb}.

The parameter λ_g is the trade-off of GIoU standard deviation for each ground truth for $\lambda_c = -0.25$. A larger λ_g indicates that more positive samples will be considered as wrong predictions. When λ_g reaches $+\infty$, the GIoU threshold is $+\infty$ and all positive samples are wrong predictions so that the loss of classification and bounding box regression cannot work. As summarized in Table VII, we set λ_g in $(-0.50, -0.25, 0, 0.25, 0.50)$ and we select the $\lambda_g = 0$, which is associated with the best performance both in 38.2% AP and 45.1% AP^{bb}.

We investigate 3 different coefficients and a control group without centerness. The results in Table VIII suggest that SBC outperforms models based on different coefficient both in AP and AP^{bb}.

We further analyze the distribution of targets in iSAID. Fig. 7(a) shows the number of regression targets for each ground truth and (b) shows the percentage of positive samples distribution for different FPN levels. Ground truth with less samples and in low FPN level are small objects. It suggests that small targets are a majority category in iSAID. Fig. 7(c) shows the deviation between centerness and GIoU distribution, which leads to the unfairness by using centerness to evaluate the quality of bounding box prediction. Therefore, it calls for

TABLE V
SUMMARY OF THE HYPERPARAMETERS OF BLENDMASK SETTINGS USED IN THIS ARTICLE

Part	Hyper-parameters	Setting Details
FCOS	INFERENCE_TH_TRAIN	0.05
	INFERENCE_TH_TEST	0.05
	LOC_LOSS_TYPE	giou
	LOSS_ALPHA	0.25
	LOSS_GAMMA	2
	NMS_TH	0.6/0.5
	POST_NMS_TOPK_TRAIN	1000/100
	POST_NMS_TOPK_TEST	1000/100
	POS_RADIUS (r)	1.5
BlendMask	ATTN_SIZE (M)	14
	BOTTOM_RESOLUTION (R)	56
	NUM_BASES (K)	4
	INSTANCE_LOSS_WEIGHT	1
	POOLER_SAMPLING_RATIO	1
	POOLER_SCALES	0.25
	POOLER_TYPE	RoIAlignV2
SOLVER	BASE_LR	0.025/0.01
	BIAS_LR_FACTOR	1
	GAMMA	0.1
	LR_SCHEDULER_NAME	WarmupMultiStepLR
	WARM_FACTOR	0.001
	WARM_ITERS	1000
	MOMENTUM	0.9

TABLE VI
EXPERIMENT TO DETERMINE THE WEIGHT OF CENTERNESS STANDARD DEVIATION IN λ_c

λ_c	AP	AP_{50}	AP^{bb}	AP^{bb}_{50}
-0.50	37.6	62.2	44.4	66.4
-0.25	**38.0**	62.6	**44.9**	66.6
0	37.8	**62.8**	44.5	66.7
0.25	37.7	62.6	44.7	**66.9**
0.50	37.6	62.4	44.6	66.6

TABLE VII
EXPERIMENT TO DETERMINE THE WEIGHT OF GIoU STANDARD DEVIATION IN λ_g

λ_g	AP	AP_{50}	AP^{bb}	AP^{bb}_{50}
-0.50	38.0	62.9	44.8	67.0
-0.25	37.9	62.7	44.4	66.6
0	**38.2**	62.7	**45.1**	66.8
0.25	38.1	**63.2**	44.7	**67.1**
0.50	37.8	62.8	44.6	66.8

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TABLE VIII
EXPERIMENT TO ASSESS THE EFFECTS OF DIFFERENT COEFFICIENTS IN CENTERNESS

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{bb}_S	AP^{bb}_M	AP^{bb}_L
w/o centerness	37.5	62.4	38.6	39.3	51.6	54.3	44.3	66.7	47.7	46.1	56.0	**77.9**
w/ centerness	37.8	62.3	38.8	39.5	**53.5**	**59.6**	44.8	66.3	**48.5**	46.7	**58.5**	76.0
ATSS [41](Our impl.)	38.0	62.2	**39.6**	39.9	52.1	54.3	44.8	66.4	48.3	46.6	57.3	76.2
SBC	**38.2**	**62.7**	39.5	**40.1**	52.0	55.7	**45.1**	**66.8**	**48.5**	**46.8**	56.8	70.5

TABLE IX
ABLATION EXPERIMENT IN CBM

w/ SAM	w/ CS	AP	AP_S	AP_M	AP_L	Time (ms)
		38.2	40.1	52.0	55.7	**74.0**
√		38.3	39.9	**53.9**	62.0	74.5
	√	38.4	40.1	53.0	**66.1**	74.8
√	√	**38.8**	**40.8**	52.8	38.6	75.6

TABLE X
EXPERIMENTS WITH VARIATION IN THE NUMBER OF CHANNELS K

K	AP	params (M)	GFLOPs (G)	Time (ms)
2	37.9	**32.15**	**93.16**	**71.8**
3	38.0	34.31	98.78	72.4
4	**38.2**	36.46	104.39	73.0
5	**38.2**	38.62	110.01	73.8

TABLE XI
EXPERIMENTS WITH CHANGES IN THE ATTENTION SHORT SIZE L

Method	L $attn$	AP	params (M)	GFLOPs (G)	Time (ms)
origin	14 784	38.2	36.46	104.39	73.0
DA-light	2 112	38.0	**29.08**	**85.15**	**70.4**
	4 224	38.0	30.31	88.36	71.1
	6 336	**38.1**	31.54	91.57	71.9
	8 448	37.9	32.77	94.78	72.6

TABLE XII
$L - M - R$ COMBINATION IN FIXED ATTENTION SIZE

Method	L M R $attn$	AP	params (M)	GFLOPs (G)	Time (ms)
origin	14 14 56 784	38.2	36.46	104.39	73.0
DA	6 28 112	**38.7**	35.23	101.19	**72.9**
	4 42 112	38.1			74.1
	4 42 168 672	38.3			77.5
	2 84 112	38.6			72.4
	2 84 168	38.4			75.8

the introduction of the SBC method to adapt the weight for extreme samples and small targets.

2) Ablation Analysis in CBM: CBM contains SAM and CS. We analyze the combination of different modules for AP and inference pure computing time. Table IX indicates that the combination of two modules leads to a higher increase in the precision compared to the sum of individual improvements obtained by each module. When combining the two modules, it improves AP from 38.2% to 38.8%.

SAM is used to locate important information in channel by computing maximum and average pooling. CS selects K channels in $T \times K$ channels for each category of targets. The CBM using both modules on channels shows better performance compared to the results obtained by summation of both modules.

3) Analysis in DA-Blender Module: In this section, we carry out analysis to finetune the hyperparameters for the DA-Blender module. The results in Table X show that the optimal number of channels K for attention and feature representation of RoI is $K = 4$, where the precision reaches saturation. Consequently, we set $K = 4$ in subsequent experiments for decomposed attention.

For short attention size L (Table XI), we keep a relatively long size of attention $M = 14$ and interpolation resolution $R = 56$, where we calculate attention size as attn $= K \times M \times L$. In Tables XI and XII, the origin shows result of original BlendMask without DA.

We investigate the influence of different combination of $L - M - R$. The results in Table XII suggest that larger M and R can be set due to space compression obtained via DA. We found that for $L = 6$, the predictive performance (AP) does not improve. Therefore, we set $L \leq 6$. In addition, we follow the default setting of ratio in [26] for $(R/M) \leq 4$. We set a maximum value of $R = 168$ since the inference time increases considerably with the increment of R. Table XII summarizes that for the set $L - M - R = 6 - 28 - 112$, we obtained a predictive performance (AP) of 38.7% when the computation cost is reduced compared to the origin approach. It proves that using redundant space to set larger M and R can improve the performance of BlendMask effectively.

Fig. 8 shows the instance generation process of BlendMask with DA ($L - M - R = 2 - 14 - 56$). Contours of all RoIs are illustrated in Fig. 8(b) and (c). While we use m-to-n mapping with m smaller than n, we can still precisely infer the details of instance by pixel despite a blurry representation of the attentions. This result suggests that attention is likely to play an auxiliary supervision role in the mask generation. Thus, it means that it is beneficial to reduce the computation costs associated with attention using a matrix decomposition.

4) Comparison With Benchmark in iSAID: Tables XIII to XVI list the results obtained in the experiments using iSAID. The results suggest that the backbone of ResNet-50, our DB-BlendMask with SBC, CBM, and DA outperforms the other methods. Tables XIII and XIV indicate a predictive performance improvement compared to Mask R-CNN of 2.5% (AP) and 3.8% (AP^{bb}). DB-BlendMask shows a relatively marginal improvement compared to BlendMask, of 1.0% (AP) and 0.9% (AP^{bb}). Fig. 9 shows the results of PRCs and AP versus IoU with backbone ResNet-50. The area under the PRCs of DB-BlendMask is larger compared to the other models, which suggests a better predictive performance. Fig. 10 provides further details on part of instance segmentation result with ResNet-50. DB-BlendMask can accurately distinguish positive samples and obtain high-quality masks. Tables XV and XVI

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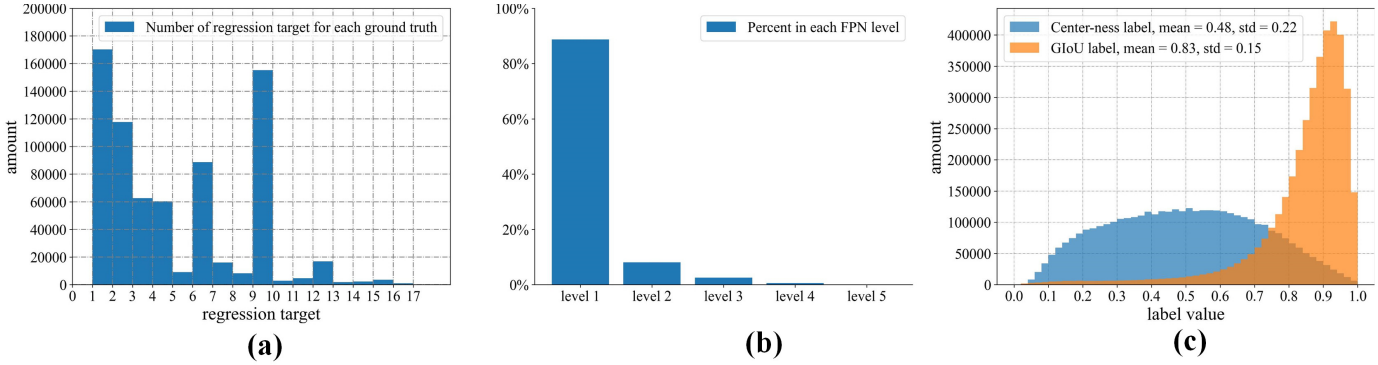


Fig. 7. Distribution of positive samples in iSAID training set with BlendMask (ResNet-50). (a) Histogram of bounding box regression of SBC over all samples. (b) Percentage distribution of positive samples in each FPN level. (c) Label distributions over all positive training samples with SBC on iSAID.

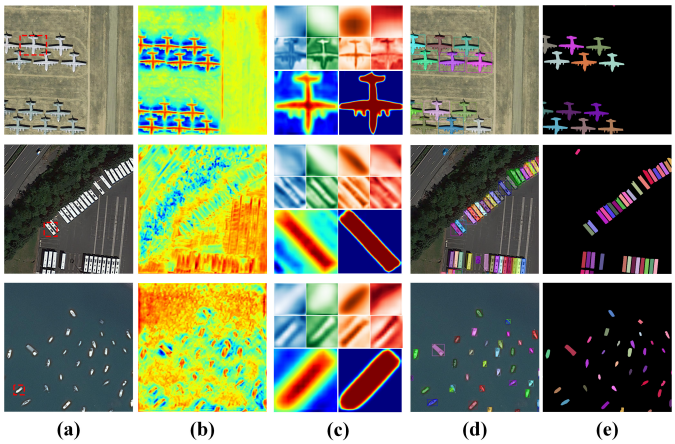


Fig. 8. Visualization of instance generation process in different scenes based on iSAID by DB-BlendMask. The rows represent scenes in an airport, a park, and at sea, respectively. (a) Original images; (b) Heat map of feature B; (c) Visualization of attention and RoI selected by red dotted box from (a) when $K = 4$. It composes of: attention with each color representing corresponding attention (first row), heat map of RoI feature with each color extracting corresponding feature of target (second row divided into), summation of attentions and RoIs (left bottom), and sigmoid summation (right bottom); (d) Detection result; (e) Ground truth.

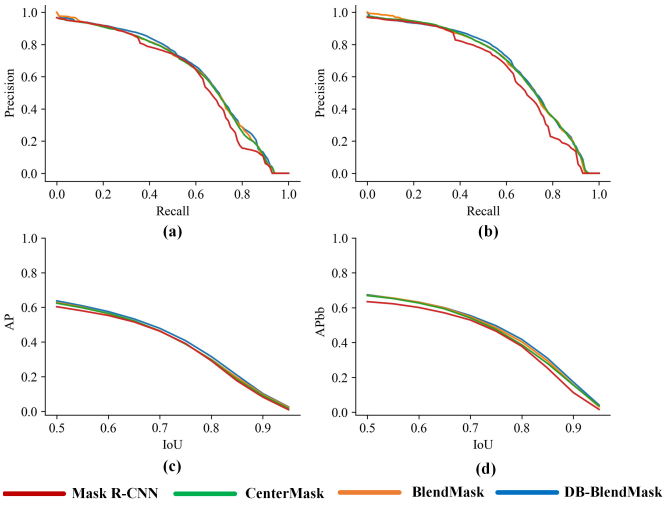


Fig. 9. PRCs and AP versus IoU in iSAID validation set. (a) PRC in AP. (b) PRC in AP^{bb}. (c) AP versus IoU. (d) AP^{bb} versus IoU.

summarize a better predictive performance (AP and AP^{bb}) for each investigated category.

The results show that Mask R-CNN based on ResNet-101 performs better compared to its implementation on ResNet-50 when FCOS-based instance segmentation model performs in contrast. Mask R-CNN uses two-stage detectors having twice positive sample selection and NMS. Therefore, additional parameters can enhance the ability of feature extraction and improve the performance by detecting more samples that are not detected in ResNet-50. FCOS is a one-stage detector with a smaller number of parameters used in the detection stage and NMS once. The enhancement of feature extraction by ResNet-101 may lead to over fitting. Furthermore, BlendMask-light appears to outperform Mask R-CNN both in both accuracy and speed. It shows an improvement of 1.7% in AP and 3.0% in AP^{bb} with 2.4 ms faster computation than Mask R-CNN.

5) Assessment of the Effects of Changes in the Instance Number: We assess effects on inference time based on changes in the instance number operated at different stages during the modeling process of ResNet-50. Instance segmentation contains feature extraction, object detection, and mask generation [Fig. 11(a)–(c)]. During the feature extraction stage, since backbone does not vary and is not related to the instance, inference time does not vary with an increase in the number of instances. In the object detection stage, both BlendMask and DB-BlendMask need to compute the additional attention. Therefore, the inference time is slightly longer than in algorithms without attention such as Mask R-CNN and CenterMask. In the mask generation stage, the number of instances after NMS varys across the models. Inference time of Mask R-CNN and CenterMask increases considerably with an increase in the number of instances since FCN is used to generate the mask. Here, the computation from the blender module is more efficient since it executes matrix operations instead of convolution. By benefiting from GPU parallel computing, the computational speed remains almost constant for blender-based models. Fig. 11(d) shows the total inference time versus instance number. For a number of instances larger

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TABLE XIII
INSTANCE SEGMENTATION RESULTS USING MASK AP ON THE iSAID TEST SET

Method	Backbone	SBC	CBM	DA	Time (ms)	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Mask R-CNN [13]			-		71.9	36.69	59.65	39.72	39.47	45.06	11.48
CenterMask [48]#			-		104.0	37.41	61.40	39.35	40.24	43.55	14.96
BlendMask [26]*					74.1	38.17	62.42	39.98	40.60	46.93	17.97
BlendMask		√			74.0	38.52	62.37	40.40	41.14	46.77	17.26
BlendMask*	R-50	√	√		75.2	38.85	63.37	40.91	41.74	**48.34**	15.72
BlendMask-light*		√		√	**69.5**	38.42	62.54	40.38	41.33	46.33	17.09
BlendMask		√		√	74.0	39.10	63.47	41.11	41.83	47.93	**21.40**
DB-BlendMask*		√	√	√	78.5	**39.21**	**63.54**	**41.41**	**42.02**	46.50	16.27
Mask R-CNN (iSAID)			-		-	33.41	56.77	34.66	35.83	46.50	**23.93**
Mask R-CNN			-		**87.9**	38.25	61.72	40.93	41.04	**48.35**	13.35
CenterMask#	R-101		-		118.4	37.17	61.02	38.85	40.29	42.69	11.89
BlendMask					91.3	38.25	62.50	40.00	**48.90**	47.40	17.90
DB-BlendMask		√	√	√	92.1	**38.86**	**63.08**	**40.98**	41.56	48.11	17.76

Given memory limitations of the test server, we adapted INFERENCE_TH_TEST (0.05) to obtain the results. This might have slightly reduced the predictive performance. The method ending with "#" means that the value of INFERENCE_TH_TEST is 0.15 and "*" is 0.06. BlendMask-light with DA means that we use combination of L-M-R = 2-14-56 and L-M-R = 3-28-112 in others DA. Mask R-CNN (iSAID) is a commonly-used benchmark [53]. DB-BlendMask is the original BlendMask with improvements. We implemented the model based on [58] and [59] and recorded the pure inference time and basic index AP, AP^{bb}.

TABLE XIV
OBJECT DETECTION RESULTS USING BOUNDING BOX AP ON THE iSAID TEST SET

Method	Backbone	SBC	CBM	DA	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{bb}_S	AP^{bb}_M	AP^{bb}_L
Mask R-CNN			-		42.19	63.11	47.76	44.67	49.87	18.86
CenterMask#			-		44.56	65.55	49.67	47.39	48.67	21.37
BlendMask*					45.06	66.08	50.08	47.50	50.56	21.90
BlendMask		√			45.47	66.58	50.61	48.23	50.94	22.28
BlendMask*	R-50	√	√		45.69	**66.79**	50.62	48.60	**52.44**	19.71
BlendMask-light*		√		√	45.22	66.36	49.98	48.17	50.28	24.86
BlendMask		√		√	**45.96**	66.68	**50.92**	48.67	51.85	**28.86**
DB-BlendMask*		√	√	√	**45.96**	66.78	50.81	**48.75**	50.67	21.15
Mask R-CNN (iSAID)			-		37.18	60.79	40.67	39.84	43.72	16.01
Mask R-CNN			-		43.93	65.48	49.66	46.44	**53.39**	18.76
CenterMask#	R-101		-		44.38	65.57	49.15	47.46	48.18	17.25
BlendMask					45.16	65.65	50.08	47.87	51.63	**22.85**
DB-BlendMask		√	√	√	**46.00**	**66.77**	**50.81**	**48.79**	52.27	22.81

TABLE XV
CLASS-WISE INSTANCE SEGMENTATION RESULTS ON THE iSAID TEST SET

Method	Backbone	AP	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
Mask R-CNN		36.7	42.9	48.0	17.4	**28.1**	15.6	30.4	46.8	75.7	48.2	35.4	30.9	43.2	31.0	38.1	**13.4**
CenterMask#		37.4	42.2	54.0	16.7	23.6	16.1	32.5	46.9	76.7	50.5	37.7	35.9	41.5	29.9	38.1	10.0
BlendMask*	R-50	38.2	43.8	55.5	15.9	21.3	15.5	33.2	46.9	76.4	50.9	37.4	**38.6**	**45.3**	33.5	**38.4**	11.0
DB-BlendMask*		**39.2**	44.7	**56.3**	**17.6**	23.7	**16.4**	**33.6**	**48.6**	**78.3**	**52.2**	**38.2**	36.9	**44.7**	**37.4**	38.1	12.4
Mask R-CNN (iSAID)		33.4	41.7	39.6	15.2	25.9	**16.9**	30.4	**48.8**	72.9	43.1	32.0	26.7	36.0	29.6	36.7	5.6
Mask R-CNN		38.3	44.1	53.1	**18.6**	**30.4**	16.1	31.1	47.0	76.7	48.2	36.9	35.7	46.4	31.1	**39.0**	14.5
CenterMask#		37.2	41.2	54.2	16.5	25.7	15.6	32.2	46.4	75.8	50.1	36.9	34.6	42.2	29.9	37.2	10.5
BlendMask*	R-101	38.3	42.6	**56.4**	16.7	22.4	16.2	33.4	47.1	77.3	**51.2**	**38.0**	36.0	**45.5**	34.0	38.2	11.8
DB-BlendMask		**38.9**	**45.2**	54.8	17.8	23.8	16.3	**34.1**	48.6	**77.5**	**51.2**	37.3	**36.3**	44.4	**36.1**	38.4	**12.2**

than 50, blender-based models perform better than FCN-based models.

While for most cases BlendMask is faster than Mask R-CNN, the results in Table XIII show that Mask R-CNN is faster than BlendMask for a specific range of number of instances. Fig. 11 shows that the maximum number of instances in Mask R-CNN is about 420. To identify the reasons behind these results, we investigate the distribution of the number of instances for each model.

We run all models based on the iSAID validation set and evaluate the number of predicted instances for each image. We compare the distribution of the predicted number of instances between the investigated models and ground truth using violin plots (Fig. 12). The number of predicted instances (y-axis) is associated with the percentage of distribution over all positive samples. The distribution of the predicted instances for Mask R-CNN shows a relatively large variability on the x-axis and tends to predict fewer positive samples. Mask R-CNN predicts fewer RoIs after twice NMS but runs faster than FCOS-based models. However, it is also affected by missing detection in small objects. The distributions of BlendMask and our DB-BlendMask are closer to those of the ground truth, which highlights the better predictive performance of BlendMask-based methods. The distribution

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TABLE XVI
CLASS-WISE OBJECT DETECTION RESULTS ON THE iSAID TEST SET

Method	Backbone	AP^{bb}	Plane	BD	Bridge	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	Harbor	SP	HC
Mask R-CNN	R-50	42.2	62.6	47.7	21.4	**40.8**	18.2	36.9	50.4	77.2	47.6	34.9	30.5	43.8	46.5	40.3	**28.6**
CenterMask#		44.6	65.9	54.8	21.4	31.1	20.4	40.9	51.1	79.5	50.8	38.7	35.4	43.7	48.7	43.9	26.7
BlendMask*		45.1	66.1	55.6	21.0	32.2	20.5	41.3	54.5	77.9	50.4	39.1	**38.9**	45.9	49.3	44.3	26.1
DB-BlendMask*		**46.0**	**67.4**	**57.0**	**22.4**	33.7	**20.7**	**41.6**	**55.2**	**80.8**	**52.6**	**39.6**	36.6	**46.4**	**51.0**	**44.5**	27.4
Mask R-CNN (iSAID)	R-101	37.2	58.5	38.5	18.6	32.7	**20.8**	36.8	51.4	72.9	43.1	32.0	26.7	36.0	29.6	**48.8**	29.6
Mask R-CNN		43.9	63.8	52.4	**23.1**	**43.9**	18.5	37.4	50.4	77.7	47.1	35.8	35.9	46.1	48.4	41.7	**31.1**
CenterMask#		44.4	65.8	55.7	21.5	32.9	20.0	40.5	53.5	79.0	50.7	37.9	34.7	44.1	48.8	43.3	25.9
BlendMask*		45.2	67.0	55.4	21.2	33.9	**20.8**	41.0	55.3	79.5	50.0	**39.2**	37.3	43.2	50.6	44.6	25.8
DB-BlendMask		**46.0**	**68.1**	**55.9**	23.0	32.8	**20.8**	**42.1**	**55.5**	**80.0**	**51.6**	39.0	**37.6**	**46.6**	**52.4**	44.6	27.7

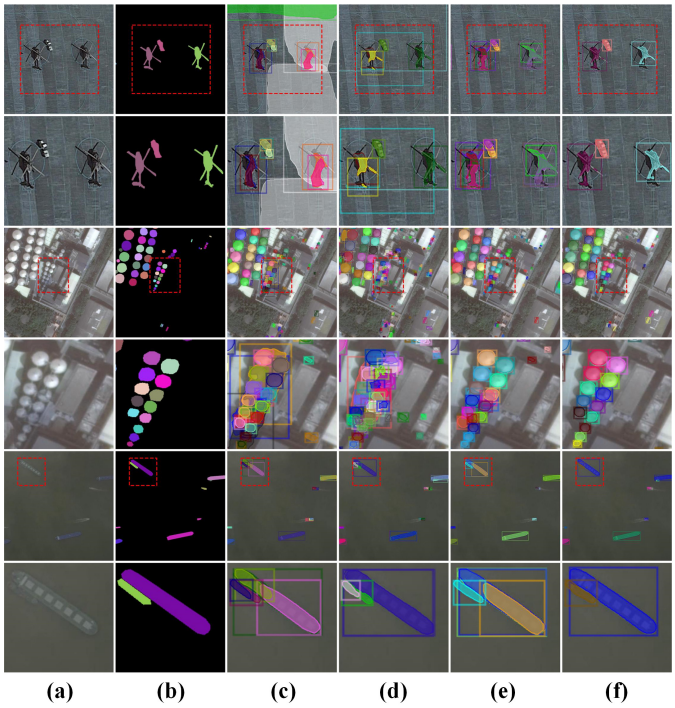


Fig. 10. Selected examples based on the iSAID validation set. Odd rows are images with original size and the enlarged area is framed with a red dotted line. Even rows are the enlarged area. (a) Original image. (b) Ground truth. (c) Mask R-CNN. (d) CenterMask. (e) BlendMask. (f) DB-BlendMask.

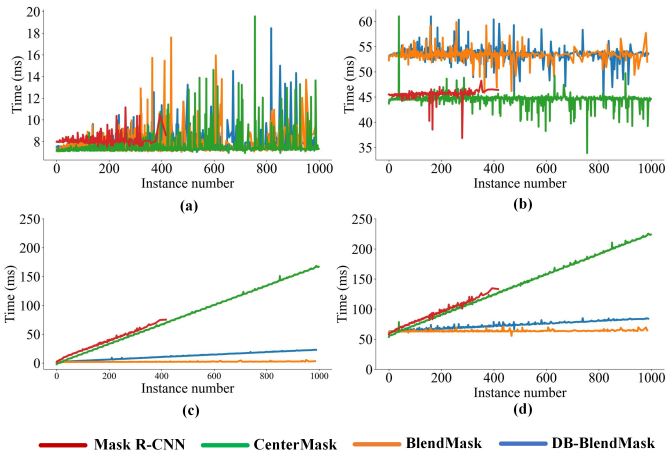


Fig. 11. Inference time versus instance number in Mask R-CNN, CenterMask, BlendMask, and DB-BlendMask. (a) Feature extraction. (b) Object detection. (c) Mask generation. (d) Total inference time.

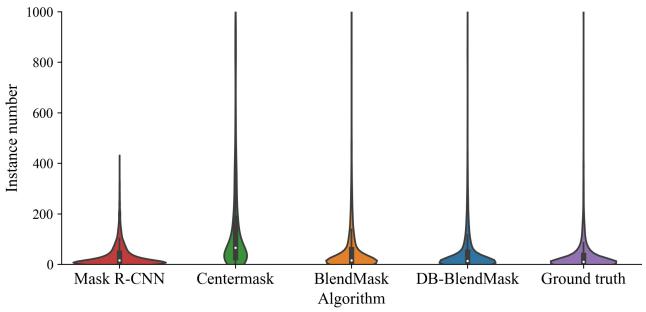


Fig. 12. Violin plots of the number of predicted instances for all investigated models and ground truth.

of the FCOS-based method CenterMask deviates from the distribution of BlendMask-based methods. Different methods of mask generation and loss contribution are likely to influence the process associated with the distribution of the predicted instances.

Fig. 13 shows the results of Mask R-CNN and DB-BlendMask for large-scale instances. Mask R-CNN exhibits a relatively poor performance in small instance detection. The area delimited by the red dotted box shows a large amount of missing detection for Mask R-CNN while DB-BlendMask can detect them all.

6) Comparison With Benchmark in ISPRS Postdam: To assess the predictive performance and computational efficiency of the proposed method in scenes with fewer instances, we compare Mask R-CNN, CenterMask, BlendMask, and DB-BlendMask based on the ISPRS Postdam test set with backbone ResNet-50. In Table XVII, we show that for a smaller detection scale, FCOS-based methods get better results both on AP and speed. DB-BlendMask reaches 63.6% AP and 66.2% AP^{bb}, which represents an improvement over Mask R-CNN of 2.7% and 3.2% (Tables XVII and XVIII). These

TABLE XVII
INSTANCE SEGMENTATION RESULTS ON THE ISPRS POSTDAM TEST SET

Method	Time (ms)	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	building	car
Mask R-CNN	86.4	60.9	83.9	69.8	17.0	50.6	41.3	57.0	64.8
CenterMask	73.6	62.3	**85.9**	72.2	17.1	51.5	41.5	59.0	**65.6**
BlendMask	**64.7**	62.0	84.1	71.5	**17.9**	51.0	**49.4**	59.4	64.6
DB-BlendMask	68.1	**63.6**	85.7	**72.3**	16.2	**51.5**	47.2	**61.5**	65.5

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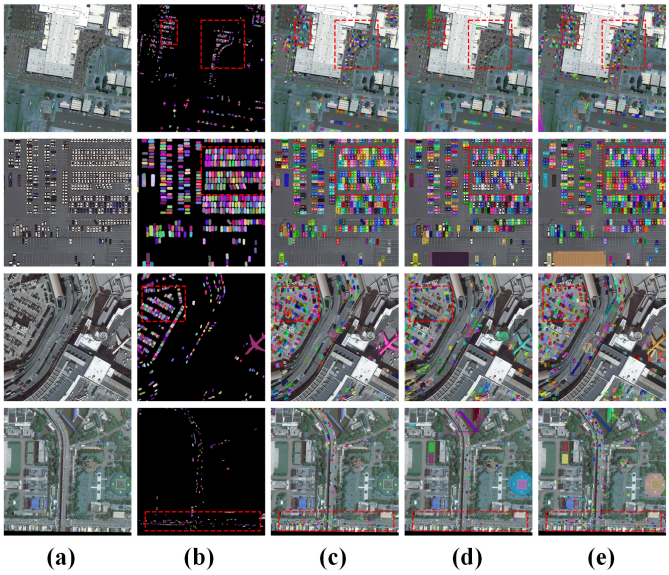


Fig. 13. Detection result in large-scale instances. The missing targets in Mask R-CNN are delimited by a red dotted line. (a) Original image. (b) Ground truth. (c) Foreground truth. (d) Mask R-CNN. (e) DB-BlendMask.

TABLE XVIII
OBJECT DETECTION RESULTS ON THE ISPRS POSTDAM TEST SET

Method	AP^{bb}	AP^{bb}_{50}	AP^{bb}_{75}	AP^{bb}_{S}	AP^{bb}_{M}	AP^{bb}_{L}	building	car
Mask R-CNN	63.0	80.1	72.6	22.1	54.7	39.7	56.6	69.3
CenterMask	65.1	**85.6**	74.2	**23.5**	55.8	40.7	59.5	70.7
BlendMask	65.0	85.5	74.4	22.2	**55.9**	40.6	59.6	70.4
DB-BlendMask	**66.2**	85.3	**75.2**	21.5	55.2	**46.1**	**61.5**	**70.8**

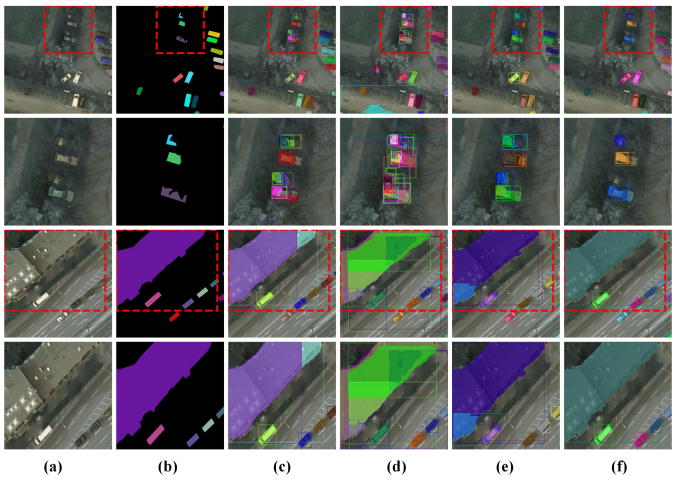


Fig. 14. Selected examples based on the ISPRS Postdam test set. Odd rows are images with original size and enlarged area is framed with a red dotted line. Even rows represent the enlarged area. (a) Original image. (b) Ground truth. (c) Mask R-CNN. (d) CenterMask. (e) BlendMask. (f) DB-BlendMask.

results suggest that our method may have a better predictive ability to detect objects in small-scale scenes. These results are illustrated in Fig. 14.

7) Failure Cases of DB-BlendMask: Fig. 15 shows the failure cases observed using our method. DB-BlendMask is sensitive to similar targets with medium size since CBM only enhances features for different categories. Furthermore, SBC is affected when the shapes of the targets are similar. This can be illustrated by a reduplication detection observed in ports (first row) and in airplanes (second row). Future work may consider setting additional adaptive rules in SBC to improve the ability of the model to distinguish objects in this particular context.

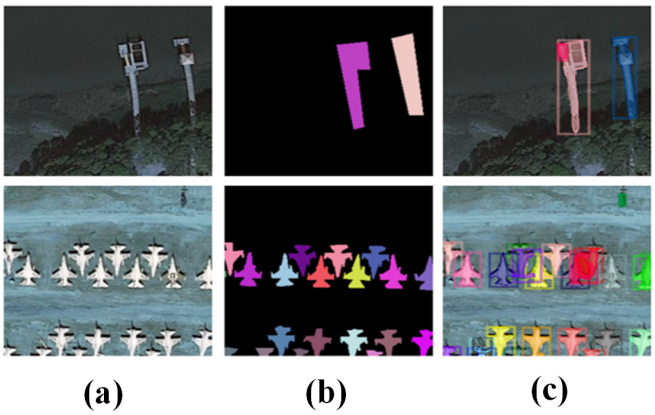


Fig. 15. Limitations of DB-BlendMask. (a) Images. (b) Ground truth. (c) Detection Results of DB-BlendMask.

V. CONCLUSION

This article proposes DB-BlendMask, a promising instance segmentation model for HRRSIs that shows high accuracy while maintaining relatively low computational costs. DB-BlendMask demonstrates its ability to account for data complexity such as 1) a high variability in both the sizes and shapes of the targets; 2) presence of a large number of small targets; and 3) data with long-tailed distribution. To address these issues, the suggested method designs SBC, which improves the performance of the weight allocation strategy for positive samples affected by a large variability in the sizes of the targets. The suggested approach guarantees that enough weights are allocated to small targets during inference and suppresses redundant predictions in large targets. Furthermore, our approach uses CBM, which combines prior knowledge obtained in object detection and spatial attention to enhance the feature representation. CBM plays an important role when targets are in minority categories. The DA-Blender module executes matrix operations, which reduce considerably the computation costs when the number of instances in HRRSIs is large. Experiments based on iSAID and ISPRS Postdam datasets indicate that DB-BlendMask has precision and computational advantages compared to the benchmark models. It achieved 39.2% AP based on iSAID and 63.6% AP based on ISPRS Postdam, which corresponds to an improvement of 2.5% and 2.7%, respectively. Future work on the optimization of the model parameters may be required to further improve the accuracy and efficiency in other contexts.

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